# Algorithmic Pseudocode Sample 3

Source-backed Image2Struct algorithm sample

## 1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

**Algorithm 2** Initialization of the control and state variables

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**for** $t \in \{-1, ..., -T^{traceback}\}$ **do** ▷ Retrieve the values of $P_{u,t}$
  $P_{u,t} \leftarrow$ `Power.GetValue`$(t)$
**end for**
**for** $t \in \{-1, ..., -T^{traceback}\}$ **do** ▷ Initial conditions on the state variables
  **if** $P_{u,t} >$ `MinimumPower.GetValue`$(t)$ **then**
    $S_{u,t}^{OFF} \leftarrow 0$
    $S_{u,t}^{STOP} \leftarrow 0$
    $S_{u,t}^{START} \leftarrow 0$
    **if** $P_{u,t} < P_{u,t-1}$ **then** ▷ Exact initialization required only if the FLAT state is defined.
      $S_{t-1}^{ON_UP} \leftarrow 0$
      $S_{t-1}^{ON_DOWN} \leftarrow 1$
      $S_{t-1}^{ON_FLAT} \leftarrow 0$
    **else if** $P_{u,t} > P_{u,t-1}$ **then**
      $S_{t-1}^{ON_UP} \leftarrow 1$
      $S_{t-1}^{ON_DOWN} \leftarrow 0$
      $S_{t-1}^{ON_FLAT} \leftarrow 0$
    **else if** $P_{u,t} = P_{u,t-1}$ **then**
      $S_{t-1}^{ON_UP} \leftarrow 0$
      $S_{t-1}^{ON_DOWN} \leftarrow 0$
      $S_{t-1}^{ON_FLAT} \leftarrow 1$
    **end if**
  **else if** $P_{u,t} > 0$ **then** ▷ Reconstruct the startups and shutdowns
    **if** $P_{u,t} < P_{u,t-1}$ **then**
      $S_t^{STOP} \leftarrow 1$
      $S_t^{START} \leftarrow 0$
    **else**
      $S_t^{STOP} \leftarrow 0$
      $S_t^{START} \leftarrow 1$
    **end if**
  **else** ▷ Final possibility: the unit is OFF.
    $S_{u,t}^{OFF} \leftarrow 1$
    $S_{u,t}^{STOP} \leftarrow 0$
    $S_{u,t}^{START} \leftarrow 0$
    $S_{t-1}^{ON_UP} \leftarrow 0$
    $S_{t-1}^{ON_DOWN} \leftarrow 0$
    $S_{t-1}^{ON_FLAT} \leftarrow 0$
  **end if**
**end for**

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